

KAISONG ZHANG

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EDUCATION

Fuzhou University | GPA: 3.93/4.0
B.S. | Software Engineering

2024 – 2028
Fuzhou, China

PUBLICATIONS

MedRQ, Second Author

Propose an ontology-driven residual vector quantization framework for drug recommendation.

- Align discrete codes with multi-level clinical ontologies.
- Generate compact and ontology-consistent medical IDs.
- Reduce the memory usage while maintaining higher accuracy than baselines.

Lv, H., **Zhang, K.**, Tan, Y., and Chen, X, “Towards Efficient and Interpretable Medical Concept Representation via Ontology-driven Residual Vector Quantization,” in *International World Wide Web Conference (WWW 2026, Short Track)*.

SHARK, First Author

Propose a disease tokenizer that models the disease representation by capturing the semantic hierarchy.

- Use Residual K-Means to recursively extracting semantic features from disease embeddings.
- Capture diseases’ convergence factors compared to the ICD taxonomy.

Zhang, K., Lv, H., Tan, Y., Lin, Z., Zhang, H., Cheng, X., and Yang, C, “SHARK: Modeling Semantic Hierarchy of Medical Code via Residual K-Means quantization,” in *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026)*.

RQCare, Second Author

Propose a Residual Quantization and Dual-Graph Structure for diagnosis prediction.

- Adopt RQ-VAE to model the hierarchical structure of clinical concepts.
- Establish patient-disease graph and disease structure graph to capture real world clinical patterns.

Yu, X., **Zhang, K.**, Lv, H., Ma, G., Lin, Z., Chen, X., and Tan, Y, “RQCare: A Residual Quantization Model for Disease Representation and Diagnosis Prediction in Healthcare Data,” in *IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2025)*.

PROJECT

Reproduce MiniOneRec

Fully reproduce MiniOneRec, a generative recommender system framework, and verify the scaling law on $3 \times A800$.

- Data are based on Amazon Reviews 18, Industrial and Scientific category, from 2016.10 to 2018.10.
- Residual quantization: encode each item’s embedding into 3-layer IDs by using RQVAE.
- SFT: extend the Qwen2.5’s vocabulary with SIDs and do multi-task fine-tuning.
- RL with 16 constrained beam search rollouts per prompt, which is optimised by GRPO.
- Analyze the theoretical GPU memory usage and compare with the practical usage.
- Verify the scaling law based on Qwen2.5-0.5B/1.5B and evaluated the $HR@\{3,5,10\}$ and $NDCG\{3,5,10\}$.

RESEARCH INTERESTS

- Data Mining
- AI For Healthcare (EHR Prediction)
- Recommender Systems
- Graph Mining